

ANALYTICAL WORKFLOW ARCHITECTURE

Real Estate Sales & Revenue Optimization

End-to-End Data Pipeline Flowchart

STAGE 01 Data Ingestion & Integration

Source Datasets

Ingest raw transactional records, property metadata (type, sqft, listing price), and customer demographic data into the analysis environment.



STAGE 02 Data Cleaning & Transformation

Python / Pandas / NumPy

Handle missing values, enforce data types, filter out outliers, and compute calculated fields (discount amounts, net selling prices, and price/sqft ratios).



STAGE 03 Exploratory Data Analysis (EDA)

Aggregation & Grouping

Aggregate monthly transaction metrics for 2026, evaluate property type performance distributions, and assess discounting impact across categories.



STAGE 04 Statistical & Regression Modeling

Seaborn / Regplot

Model the structural correlation between property square footage (`area_sqft`) and final selling price (`sale_price`) to evaluate dynamic pricing baselines.

◆ KEY METRICS & FINDINGS UNCOVERED

- **Total Revenue:** £8,000,000 across 22 units
- **Average Sale Price:** £363,636.36 (Median: £335,000)
- **Average Discount:** £9,545.45 (~0.12% total margin ratio)
- **Top Revenue Segment:** Houses (£3,610,000 revenue)
- **Top Volume Segment:** Apartments (7 units sold)
- **Key Insight:** High price retention with linear sqft scaling

Strategic Action Plan (Output Recommendations)

1. Inventory Pivot

Focus acquisition effort on detached/semi-detached Houses to maximize overall revenue yield per sale.

2. Dynamic Pricing

Standardize baseline £/sqft pricing tiers using empirical regression metrics to optimize quotes.

3. Margin Control

Maintain tight pricing power on top categories while targeting small discounts only for slower inventory.